

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|--------------------------------|---|
| 1. Scalar D | (A) The number of independent columns of a matrix. |
| 2. Magnitude F | (B) A vector whose length is exactly 1. |
| 3. Unit vector B | (C) All vectors you can reach as $A\mathbf{x}$ — the span of a matrix's columns. |
| 4. Linear combination G | (D) A single number that scales (stretches or flips) a vector. |
| 5. Span H | (E) $\mathbf{v} \cdot \mathbf{w} = v_1w_1 + v_2w_2$; it measures length and angle. |
| 6. Dot product E | (F) The length of a vector, $\sqrt{v_1^2 + v_2^2}$. |
| 7. Column space C | (G) A sum of scaled vectors, $c\mathbf{v} + d\mathbf{w}$. |
| 8. Rank A | (H) The set of <i>all</i> linear combinations of a set of vectors. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- Two nonzero vectors are **perpendicular** exactly when (A) their sum is **0** (B) $\mathbf{v} \cdot \mathbf{w} = 0$ ← (C) $\mathbf{v} \cdot \mathbf{w} = 1$ (D) they have equal length
- The vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 6 \end{bmatrix}$ span (A) a single point (B) **a line** ← (C) the whole plane (D) all of space
- The product $A\mathbf{x}$ is (A) **a combination of the columns of A** ← (B) a combination of the rows of A (C) always **0** (D) the length of $\mathbf{x}$
- The matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has rank (A) 0 (B) **1** ← (C) 2 (D) 4
- The unit vector $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ has length (A) 0 (B) **1** ← (C) $\|\mathbf{v}\|$ (D) $\|\mathbf{v}\|^2$
- A vector $\mathbf{b}$ lies in the column space of A exactly when (A) $\mathbf{b} = \mathbf{0}$ (B) **$A\mathbf{x} = \mathbf{b}$ has a solution** ← (C) $\mathbf{b}$ is a unit vector (D) $\mathbf{b}$ is perpendicular to every column

Part C — Short Answer & Computation (30 pts)

Show your work.

1. Let $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute (a) $\mathbf{v} + \mathbf{w}$, (b) $2\mathbf{v} - \mathbf{w}$, (c) $\|\mathbf{v}\|$.

$$\begin{aligned}\mathbf{v} + \mathbf{w} &= \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix} \\ 2\mathbf{v} - \mathbf{w} &= \begin{bmatrix} 6 \\ 8 \end{bmatrix} - \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} \\ \|\mathbf{v}\| &= \sqrt{3^2 + 4^2} = \sqrt{25} = 5\end{aligned}$$

2. Write the unit vector that points the same direction as $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

$$\begin{aligned}\|\mathbf{v}\| &= 5 \\ \frac{\mathbf{v}}{\|\mathbf{v}\|} &= \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}\end{aligned}$$

3. Let $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{u} \cdot \mathbf{v}$, then state whether $\mathbf{u}$ and $\mathbf{v}$ are perpendicular and how you know.

$$\begin{aligned}\mathbf{u} \cdot \mathbf{v} &= (4)(-1) + (2)(2) \\ &= -4 + 4 = 0 \\ &\Rightarrow \text{perpendicular — the dot product is 0}\end{aligned}$$

4. Find weights c and d so that $c\begin{bmatrix} 2 \\ 1 \end{bmatrix} + d\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$.

$$\begin{aligned}2c + d &= 5 \\ c + 3d &= 5 \\ c &= 2, \quad d = 1 \\ 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} &= \begin{bmatrix} 5 \\ 5 \end{bmatrix}\end{aligned}$$

5. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Compute $A\mathbf{x}$ as a combination of the columns of A .

$$\begin{aligned}A\mathbf{x} &= 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 6 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}\end{aligned}$$

6. Factor $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$ as $A = CR$. Which columns are independent? Write C , build the recipe R , and give the rank.

$$\begin{aligned}\text{col } 3 &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \quad (\text{columns 1, 2 independent}) \\ C &= \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \\ \text{rank} &= 2\end{aligned}$$

7. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. For each target, say whether it is reachable as $A\mathbf{x}$ (is it in the column space?) and justify in one line. (a) $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ (b) $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$C(A) = \text{the line through } \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(a) $\begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ reachable

(b) $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \neq c \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ not reachable

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. A matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ has columns $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$. (a) Is the column space a line or the whole plane? How do you know? (b) Is $\mathbf{b} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ reachable? If so, find the weights. (c) If the second column were changed to $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$, what would happen to the column space, and why? (a) **The whole plane — the columns are not multiples, so they span $\mathbb{R}^2$.**

$$c \begin{bmatrix} 1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$c + 2d = 4$$

$$c + 3d = 5$$

$$d = 1, \quad c = 2$$

(c) **Column 2 becomes 2× column 1, so $C(A)$ shrinks to a line (rank 1).**

2. Two viewers' taste profiles are $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{u} \cdot \mathbf{v}$ and explain what the result says about how similar the two viewers' tastes are, using what you know about the sign of a dot product.

$$\mathbf{u} \cdot \mathbf{v} = (2)(-1) + (1)(2) = 0$$

Zero means perpendicular — neither aligned nor opposed, so tastes are unrelated. ... A positive dot would mean similar tastes; a negative dot, opposite tastes.